

Aegis Home System — Draft v2 V2.0

Adding Cold Storage and Microwave Appliances

Version 2.0 — June 2026

Iteration 2 adds refrigeration and microwave to the V1 kitchen core.

Δ Change Log (V1 → V2)

- Stellar-Chill: PMN-PT electrocaloric film COP > 4 (2025)
- Safe-Wave: GaN solid-state RF source 97% efficiency vs magnetron 65%

New V2 Components Added to V1 Base

5. **Stellar-Chill Refrigerator (CSMFAB0000000000004)** — solid-state thermoelectric/electrocaloric cooling; no HFC refrigerants; Aegis-C BFRP chassis; VARTM fabricated; hot-swappable cooling cartridge
6. **Safe-Wave Microwave (CSMFAB0000000000005)** — GaN solid-state 2.45 GHz source (no magnetron, no transformer); multi-layer Faraday containment; optically isolated ceramic viewing window; Aegis-C composite housing

V2.0 Updates for Stellar-Chill: - PMN-PT electrocaloric film replacing V1 Bi₂Te₃ modules - COP > 4 at 10°C differential (vs COP 0.3–0.6 for bulk Peltier) - PVDF-TrFE-CFE 5-layer stack, 12°C adiabatic ΔT at 50 MV/m

V2.0 Updates for Safe-Wave: - GaN HEMT RF amplifier at 2.45 GHz: 97% drain efficiency (Class-F topology) - vs. magnetron: 65% efficiency, uncontrolled spectrum, >4 kV requirement - Total Safe-Wave power consumption: 890W vs equivalent magnetron unit 1,100W

V2 cumulative GIC threat reduction: ~58% (kitchen + cold storage + microwave hardened)